

A shared language for Midflight

What the team wants > What each agent intends to build > What was actually built

TERM	DEFINITION	CHECKOUT EXAMPLE
Plan	The team's approved product direction and division of work; versioned as decisions change.	Build an online shop; Alice builds the backend and Bob builds the checkout page.
Requirement	A specific behavior the product must satisfy.	Display the total calculated by the backend.
Task	A piece of work assigned to a person and their coding agent.	Build the checkout page.
Claim	An agent's declaration of how it intends to perform a task; intention, not completion.	The page will read total as a decimal dollar amount.
Assumption	A detail relied on but not yet established by the plan, authoritative code, or an agreement.	The backend will return a field named total.
Dependency	Something one task needs from another task or existing component.	The page consumes the checkout response.
Agreement / contract	A recorded decision about how connected components must work together.	Return total_cents as an integer; convert it for display.
Discrepancy	A mismatch between connected claims, or between implementation and an approved decision.	The page expects total; the backend returns total_cents.
Directive	A scoped update to an affected agent based on an approved requirement or agreement.	Read total_cents and update the mock response.
Evidence	Observable information about the implementation.	Pushed code and tests tied to the reviewed commit.
Verification	Comparing evidence with the current requirements, claims, and agreements.	Check that the backend and page use the agreed price format.

LEGEND

Process or update

Possible mid-work change

1-8 Normal path

C1-C4 Change path

Intent

Coordination

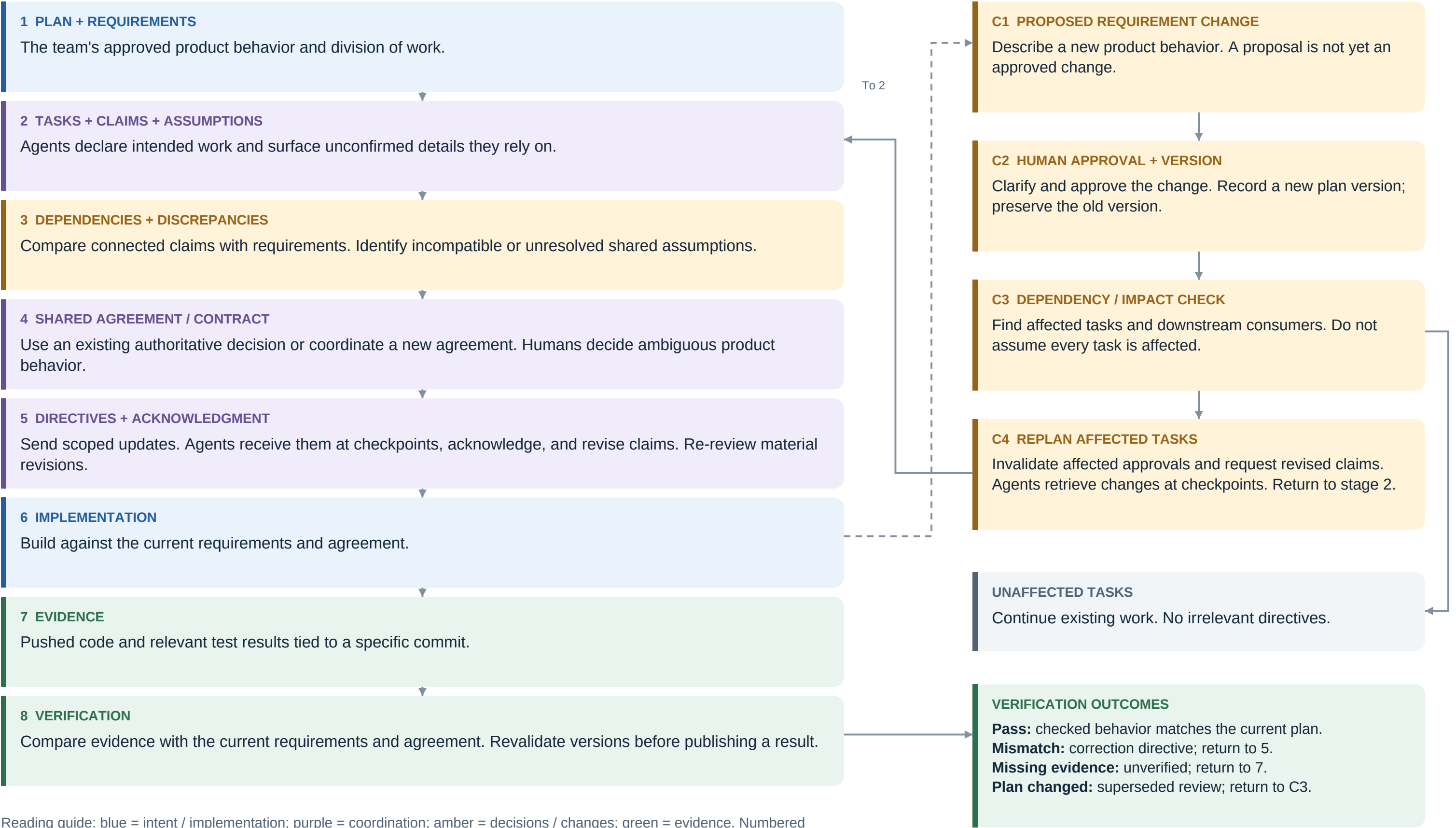
Decision / change

Evidence / verification

Assumptions are not automatically mistakes. Acknowledged does not mean implemented; verified covers only the checked behavior.

The coordination workflow, including change

Read the left column top to bottom. The right column can start at any time; only affected tasks return to planning.



Reading guide: blue = intent / implementation; purple = coordination; amber = decisions / changes; green = evidence. Numbered returns keep feedback paths readable.

Changes are delivered at checkpoints. Pending, acknowledged, implemented, and verified are separate states.

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graph TD
    S1["1 PLAN + REQUIREMENT v1  
Build a shop. The checkout page displays the final total calculated by the backend."]
    S2["2 TASKS + CLAIMS + ASSUMPTIONS  
Alice: return total_cents: 4999 (cents).  
Bob: expect total: 49.99 (dollars). Each assumes the other will match."]
    S3["3 DEPENDENCY + DISCREPANCY  
Bob's page consumes Alice's response. The proposed field names and units are incompatible."]
    S4["4 SHARED AGREEMENT / CONTRACT  
v1: integer total_cents. After the approved change, v2 adds subtotal_cents and tax_cents. Agents accept the applicable contract."]
    S5["5 DIRECTIVES + ACKNOWLEDGMENT  
v1: Bob updates his consumer. v2: Alice adds the breakdown; Bob displays it. Receive updates at checkpoints and revise claims."]
    S6["6 IMPLEMENTATION  
Both agents build against the current agreement and update relevant tests."]
    S7["7 EVIDENCE  
Pushed code plus response and page tests tied to the reviewed commit."]
    S8["8 VERIFICATION  
Does the response AND the page satisfy the current version? Code implementing only v1 cannot pass against approved v2."]

    C1["C1 PROPOSED REQUIREMENT  
Show subtotal and tax separately before the customer pays."]
    C2["C2 TEAM APPROVES PLAN v2  
Keep the final total; add the breakdown. In this demo, subtotal plus tax equals total."]
    C3["C3 DEPENDENCY / IMPACT CHECK  
Alice must provide the amounts. Bob must display them. Product-description work is unrelated."]
    C4["C4 REPLAN AFFECTED TASKS  
Revise Alice and Bob's claims for v2. Return to stage 2 with new claims, not their original assumptions."]
    U["UNAFFECTED TASKS  
Product-description work continues."]
    V["VERIFICATION OUTCOMES  
Pass: checked behavior matches the current plan.  
Mismatch: correction directive; return to 5.  
Missing evidence: unverified; return to 7.  
Plan changed: superseded review; return to C3."]

    S1 --> S2
    S2 --> S3
    S3 --> S4
    S4 --> S5
    S5 --> S6
    S6 --> S7
    S7 --> S8
    S6 -.-> C4
    C4 -- "To 2" --> S2
    S8 --> V
    C1 --> C2
    C2 --> C3
    C3 --> C4
    C3 --> U
    U --> V
  
```

1 PLAN + REQUIREMENT v1
Build a shop. The checkout page displays the final total calculated by the backend.

2 TASKS + CLAIMS + ASSUMPTIONS
Alice: return total_cents: 4999 (cents).
Bob: expect total: 49.99 (dollars). Each assumes the other will match.

3 DEPENDENCY + DISCREPANCY
Bob's page consumes Alice's response. The proposed field names and units are incompatible.

4 SHARED AGREEMENT / CONTRACT
v1: integer total_cents. After the approved change, v2 adds subtotal_cents and tax_cents. Agents accept the applicable contract.

5 DIRECTIVES + ACKNOWLEDGMENT
v1: Bob updates his consumer. v2: Alice adds the breakdown; Bob displays it. Receive updates at checkpoints and revise claims.

6 IMPLEMENTATION
Both agents build against the current agreement and update relevant tests.

7 EVIDENCE
Pushed code plus response and page tests tied to the reviewed commit.

8 VERIFICATION
Does the response AND the page satisfy the current version? Code implementing only v1 cannot pass against approved v2.

C1 PROPOSED REQUIREMENT
Show subtotal and tax separately before the customer pays.

C2 TEAM APPROVES PLAN v2
Keep the final total; add the breakdown. In this demo, subtotal plus tax equals total.

C3 DEPENDENCY / IMPACT CHECK
Alice must provide the amounts. Bob must display them. Product-description work is unrelated.

C4 REPLAN AFFECTED TASKS
Revise Alice and Bob's claims for v2. Return to stage 2 with new claims, not their original assumptions.

UNAFFECTED TASKS
Product-description work continues.

VERIFICATION OUTCOMES
Pass: checked behavior matches the current plan.
Mismatch: correction directive; return to 5.
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How to turn the diagrams into a reliable service

Implementation guidance for agents. This reference explains behavior; it does not claim the service is already built.

AUTHORITY

Humans approve product behavior and requirement changes. Agents propose technical details. Midflight coordinates agreements under the team's policy; model output cannot grant itself authority.

REAL CONNECTIONS

GitHub supplies pushed code, not unreported agent intentions. Connected agents must submit claims and consult Midflight at checkpoints. A GitHub installation alone cannot provide pre-implementation coordination.

VERSIONED STATE

Bind claims, agreements, directives, and reviews to versions. Identify the exact reviewed commit. If the plan or dependent agreement changes during review, revalidate before publishing success.

DELIVERY IS NOT COMPLETION

Track queued, acknowledged, implemented, and verified separately. A checkpoint is a cooperative integration point, not a guarantee of immediate interruption. Show when an agent has not received an update.

TARGETED CHANGES

Follow dependencies to find affected tasks. Request revised claims and re-review them before approval. Unaffected work continues. Existing agreements must not be silently replaced by new assumptions.

EVIDENCE AND FAILURE

Check code and relevant tests, not just completion messages. Missing or stale evidence remains unverified. GitHub errors suspend dependent approvals and directives until fresh data is available.

DEMO FIXTURE - BEFORE COORDINATION

Alice proposes

total_cents: 4999, currency: USD

Bob independently mocks

total: 49.99, currency: USD

Separate tests can pass; different files can merge cleanly. The combined components still disagree.

DEMO FIXTURE - APPROVED v2

subtotal_cents: 4500

tax_cents: 499

total_cents: 4999

currency: USD

The page shows subtotal \$45.00, tax \$4.99, and total \$49.99. Verify consumer behavior as well as response fields.

Demo success: catch the original mismatch; agree on v1; approve the mid-build change; update only affected tasks; reject v1-only code against v2; verify the corrected behavior. The amounts are synthetic, not tax guidance.